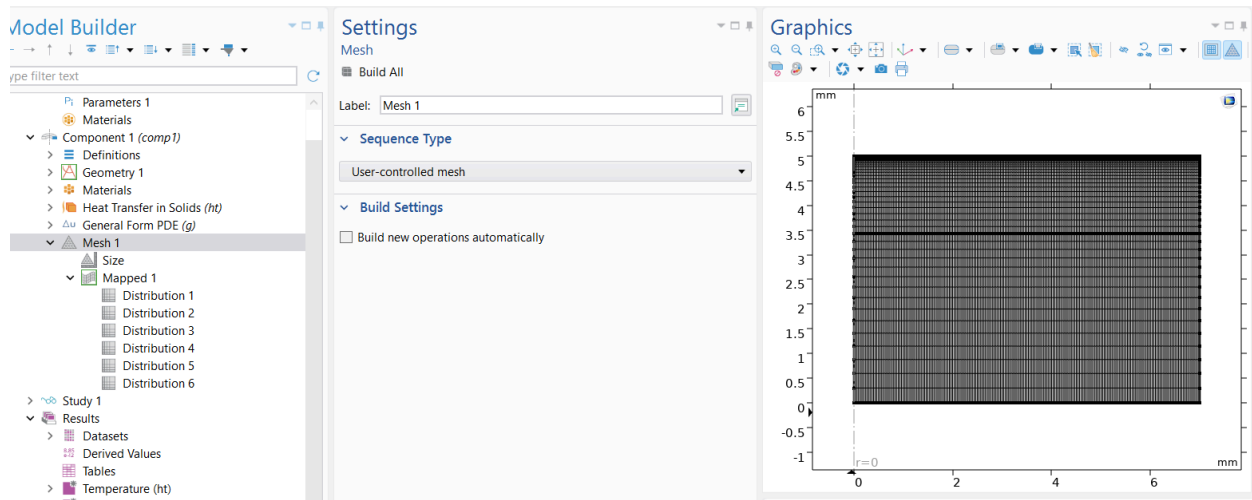
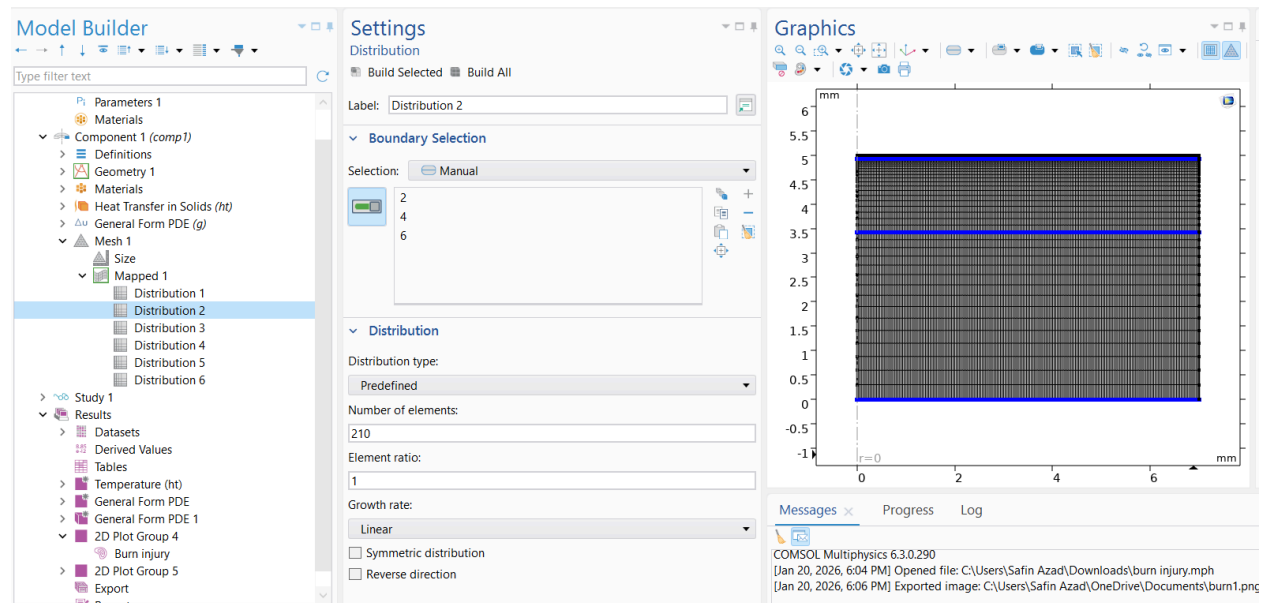
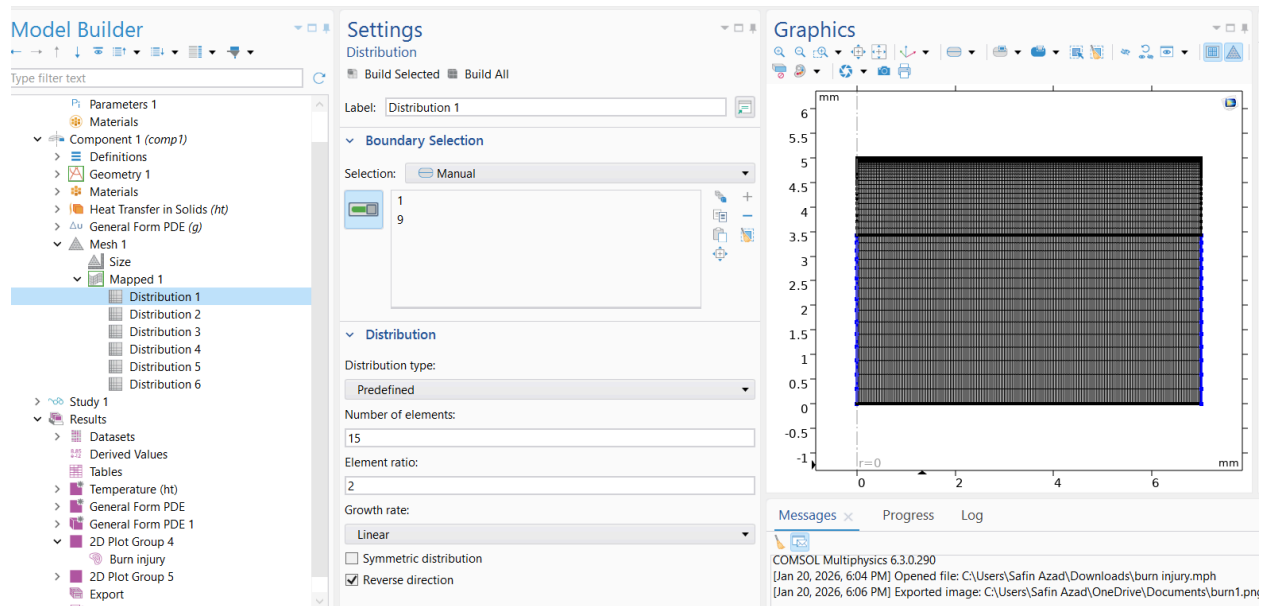


1. In your Class work you have applied mesh at different regions of the skin layers with different proportions, Now the total number of elements in the whole mesh (smallest rectangles) is 9450. Show the calculation of how the number of elements is achieved (closely look at the distributions of mesh). As you understand the process, attach necessary screenshots with calculations



*The total number of elements in the whole mesh (smallest rectangles) is 9450.



Model Builder

Type filter text

- Parameters 1
- Materials
- Component 1 (comp1)
 - Definitions
 - Geometry 1
 - Materials
 - Heat Transfer in Solids (ht)
 - General Form PDE (g)
 - Mesh 1
 - Size
 - Mapped 1
 - Distribution 1
 - Distribution 2
 - Distribution 3**
 - Distribution 4
 - Distribution 5
 - Distribution 6
- Study 1
- Results
 - Datasets
 - Derived Values
 - Tables
 - Temperature (ht)
 - General Form PDE
 - General Form PDE 1
 - 2D Plot Group 4
 - Burn injury
 - 2D Plot Group 5
 - Export

Settings

Distribution

Build Selected Build All

Label: Distribution 3

Boundary Selection

Selection: Manual

3
10

Distribution

Distribution type: Predefined

Number of elements: 20

Element ratio: 20

Growth rate: Linear

☐ Symmetric distribution

☒ Reverse direction

Graphics

Messages Progress Log

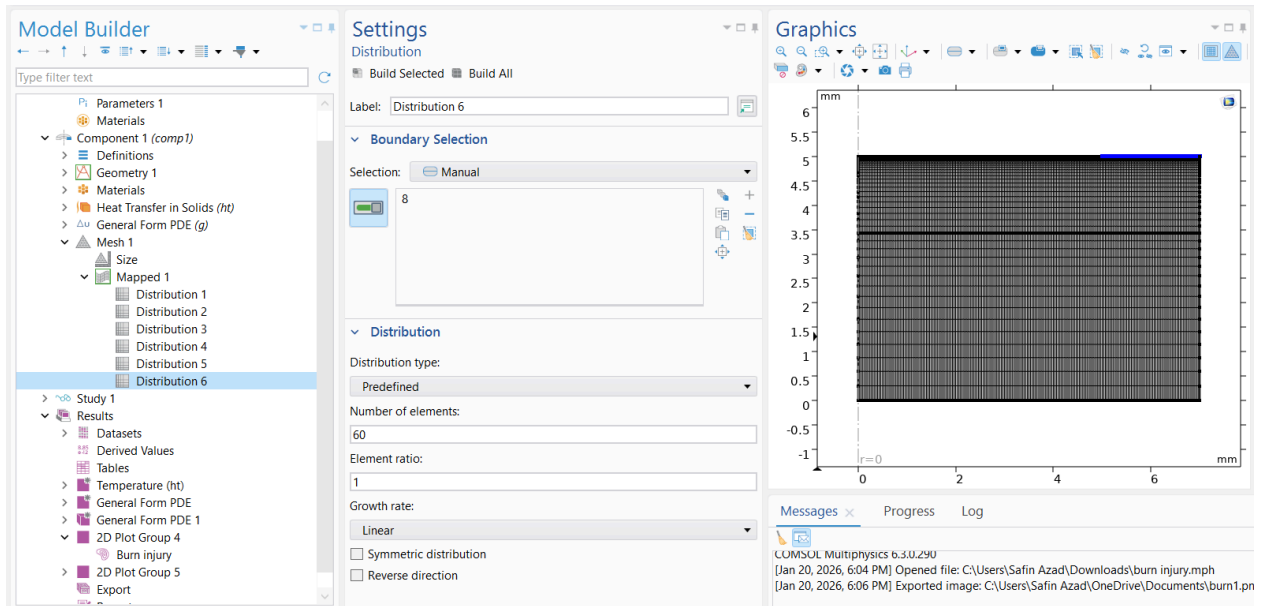
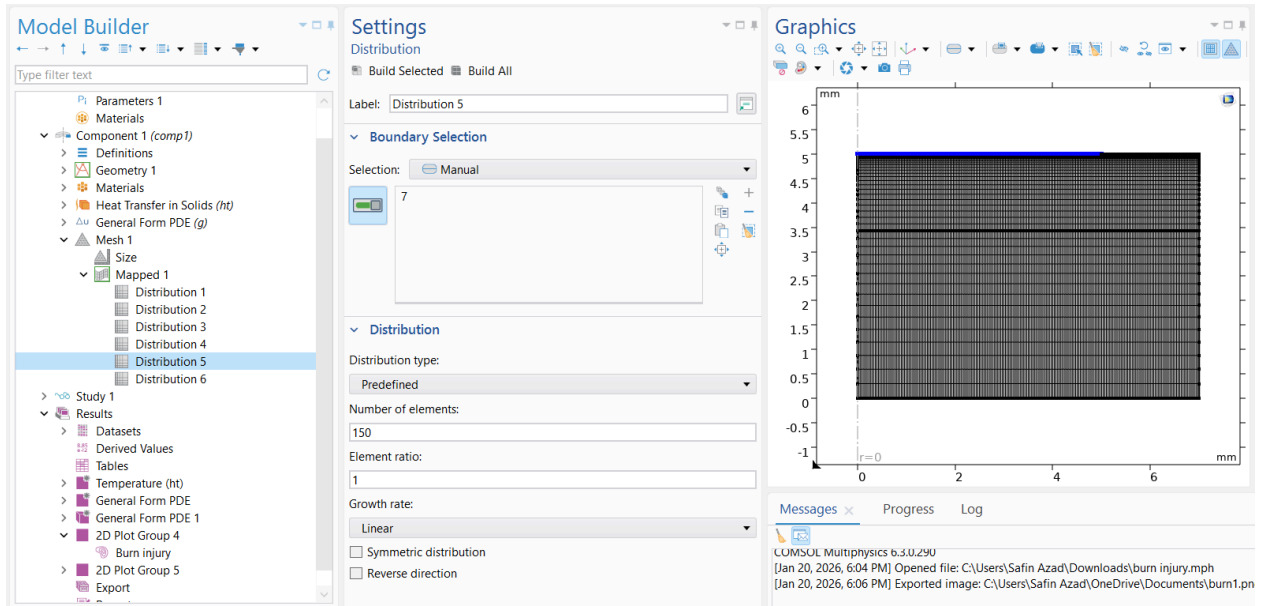
COMSOL Multiphysics 6.3.0.290

[Jan 20, 2026, 6:04 PM] Opened file: C:\Users\Safin Azad\Downloads\burn injury.mph

[Jan 20, 2026, 6:06 PM] Exported image: C:\Users\Safin Azad\OneDrive\Documents\burn1.png

The screenshot displays the COMSOL Multiphysics software interface, divided into three main panes:

- Model Builder:** Shows the hierarchical structure of the model. The 'Mesh' node is expanded, revealing 'Size' and 'Mapped 1'. Under 'Mapped 1', several 'Distribution' nodes are listed, with 'Distribution 4' selected and highlighted in blue.
- Settings:** The 'Distribution' settings for 'Distribution 4' are shown. The 'Boundary Selection' section is active, displaying a 'Manual' selection with two elements (5 and 11). The 'Distribution' section shows 'Distribution type' set to 'Predefined', 'Number of elements' set to 10, 'Element ratio' set to 1, and 'Growth rate' set to 'Linear'. There are also checkboxes for 'Symmetric distribution' and 'Reverse direction', both of which are currently unchecked.
- Graphics:** The 'Graphics' pane shows a 2D plot of the mesh. The x-axis is labeled 'mm' and ranges from 0 to 6. The y-axis is labeled 'mm' and ranges from -1 to 6. The plot shows a rectangular domain with a fine, structured mesh. A label 'r=0' is visible near the bottom-left corner of the mesh.



To find the total elements, multiply the total horizontal elements by the vertical elements for each domain: Total Horizontal Elements : $15 + 150 + 60 = 225$ elements.

Vertical Elements per Layer: Epidermis: 20 elements

Dermis: 10 elements

Subcutaneous Fat: 210 element

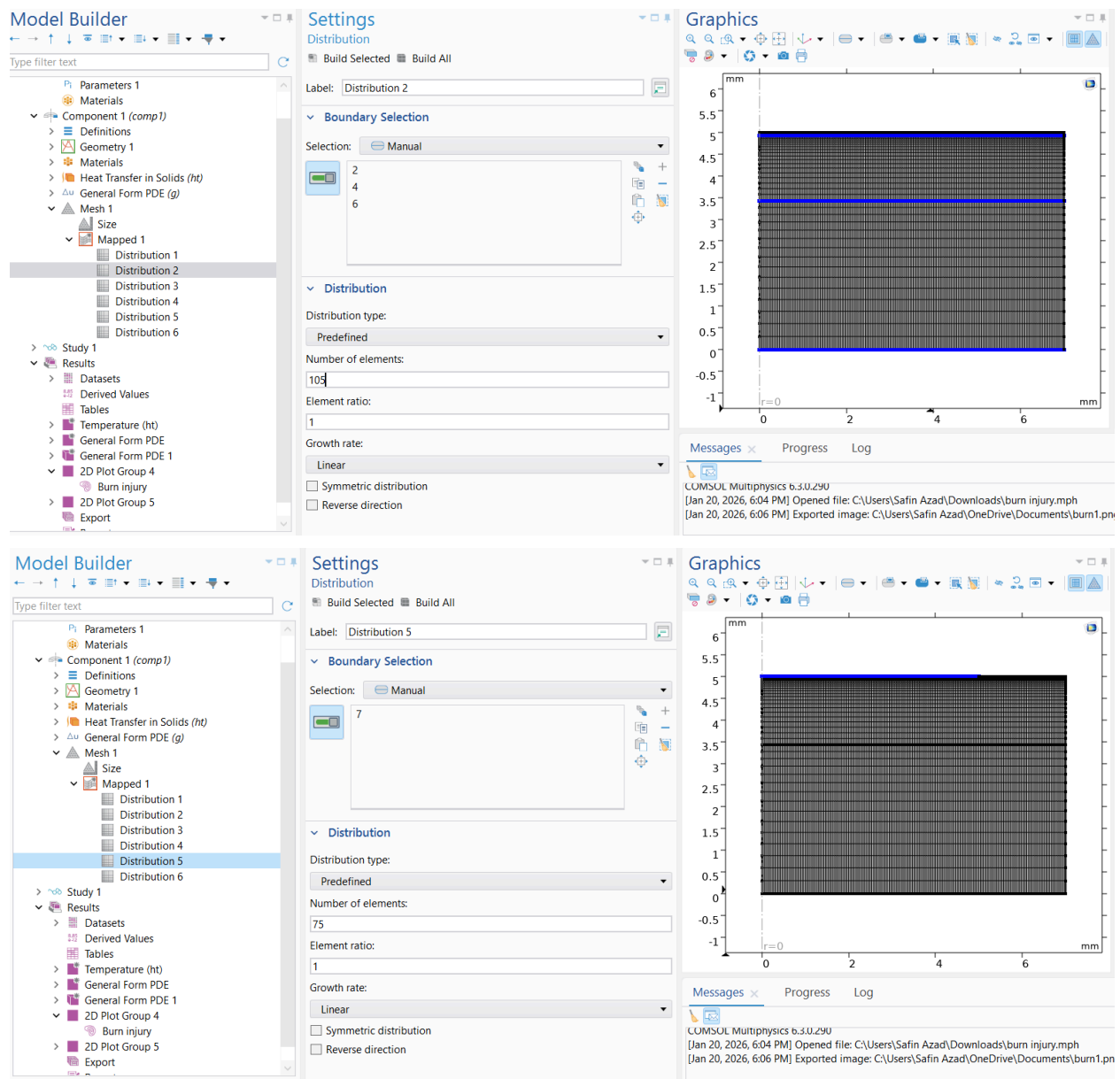
Final Total Calculation:

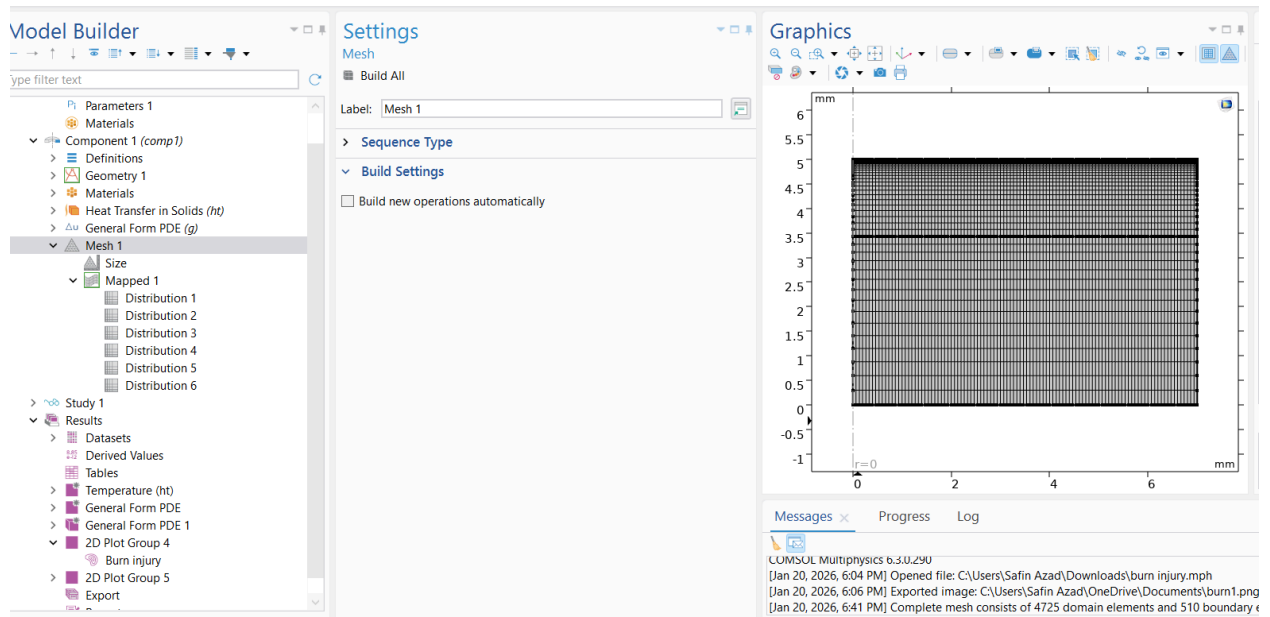
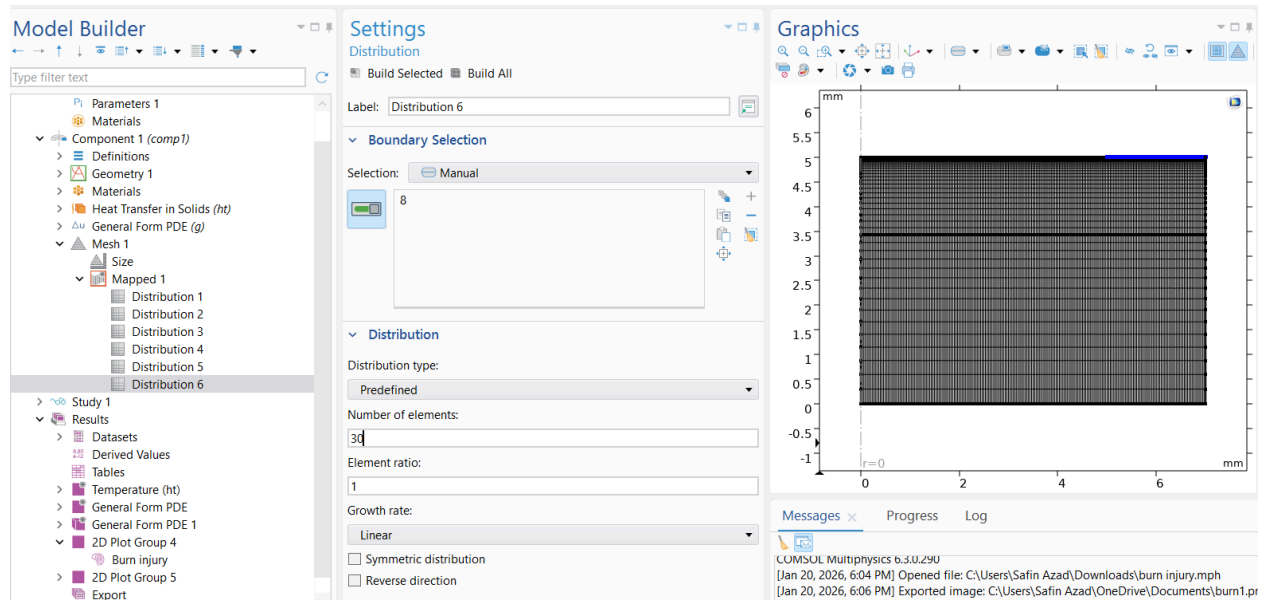
The total number of elements is the sum of elements in all three layers:

Epidermis: $225 \times 20 = 4,500$

Dermis: $225 \times 10 = 2,250$
 Subcutaneous Fat: $225 \times 30 = 2700$
 Total element is: $4,500 + 2,250 + 2,700 = 9,450$

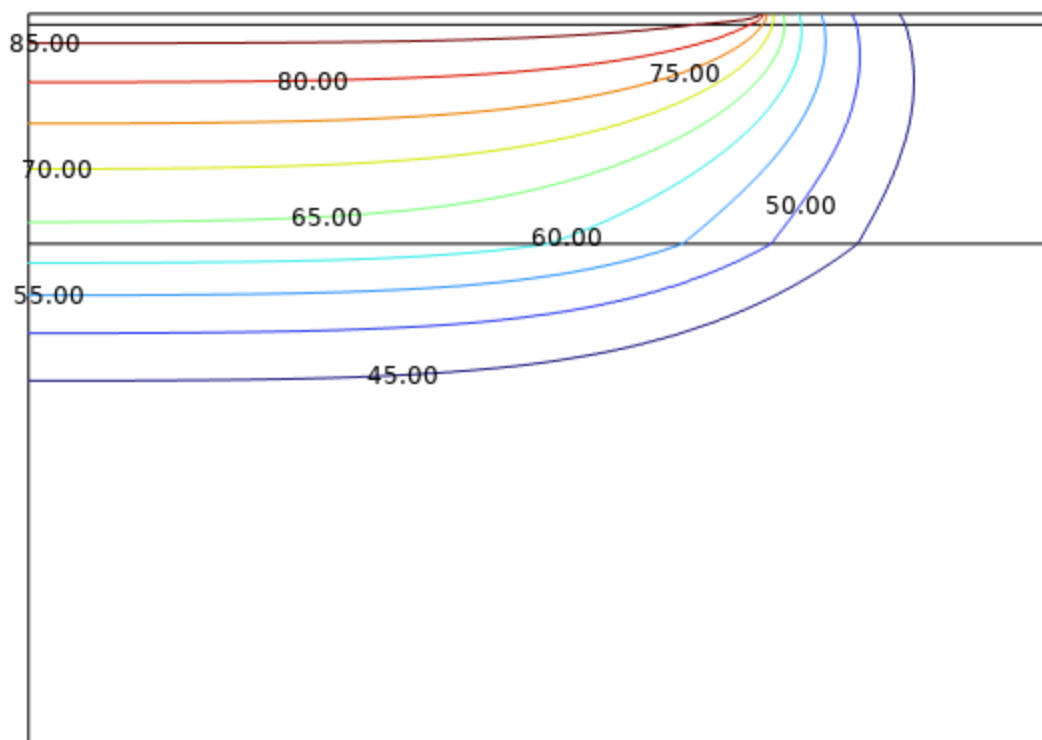
2. Now for the classwork, the total number of elements was 9450, arrange the mesh (distribution) in such a way so that the number of elements is i) half and ii) twice the base case. For both cases, show burn injury profile and temperature contour. Comment on the changes.

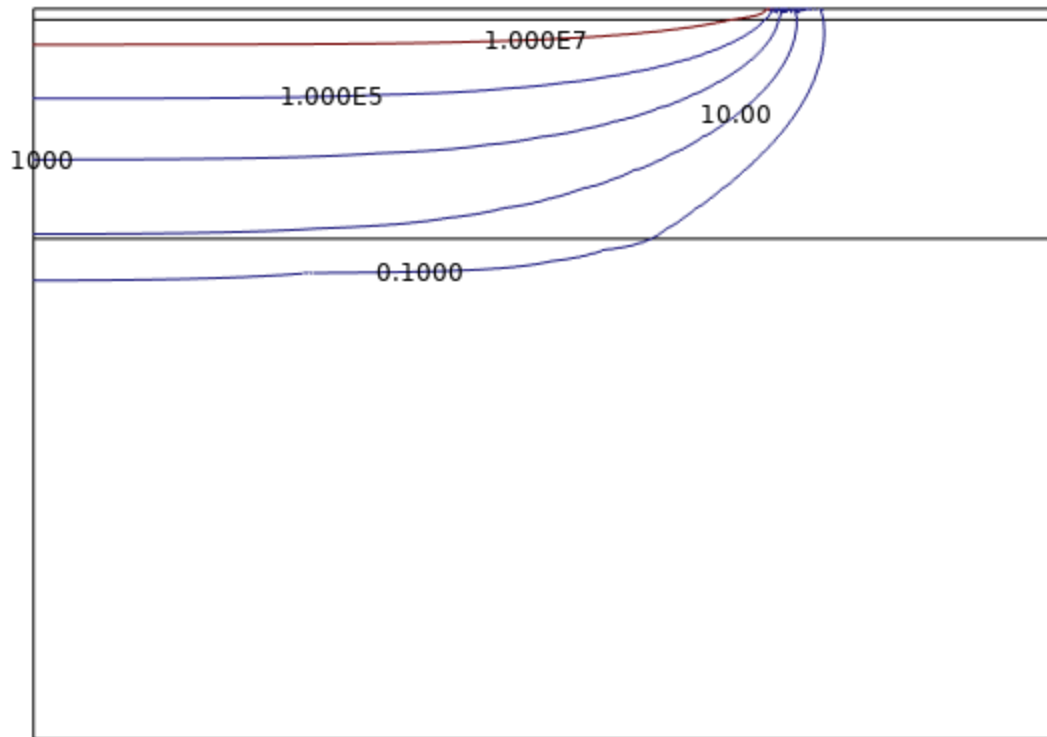




For half as base, we take distribution 2, distribution 5, distribution 6 respectively 105, 75 and 30. Thus the total element will be 4725.

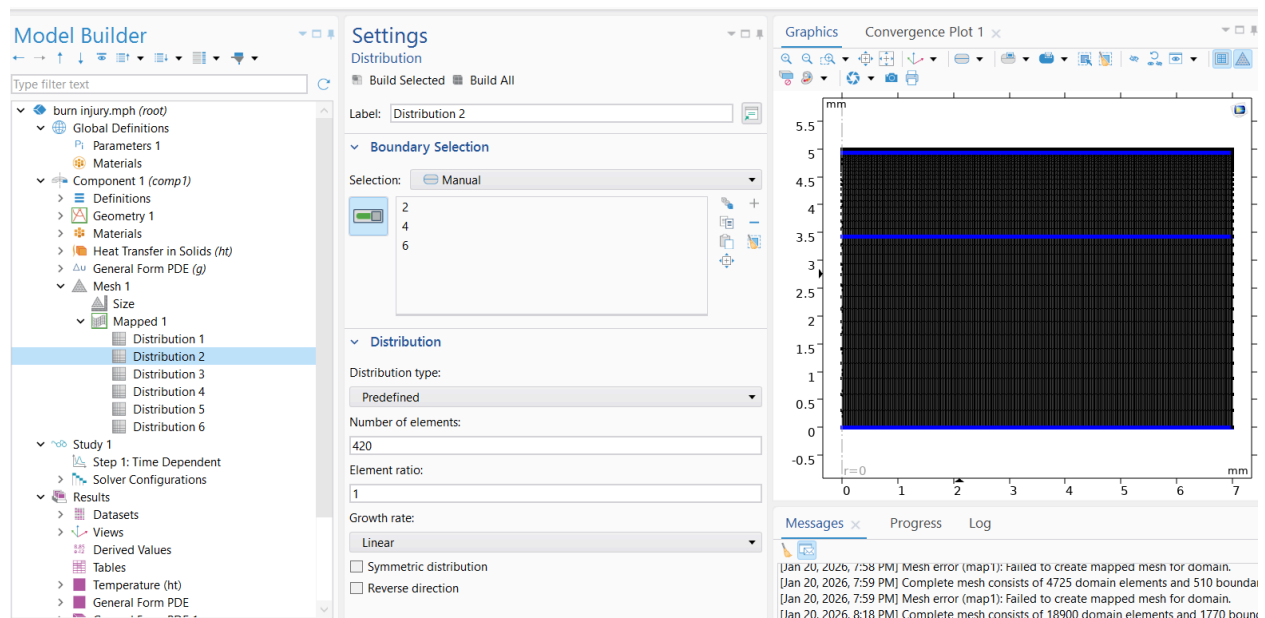
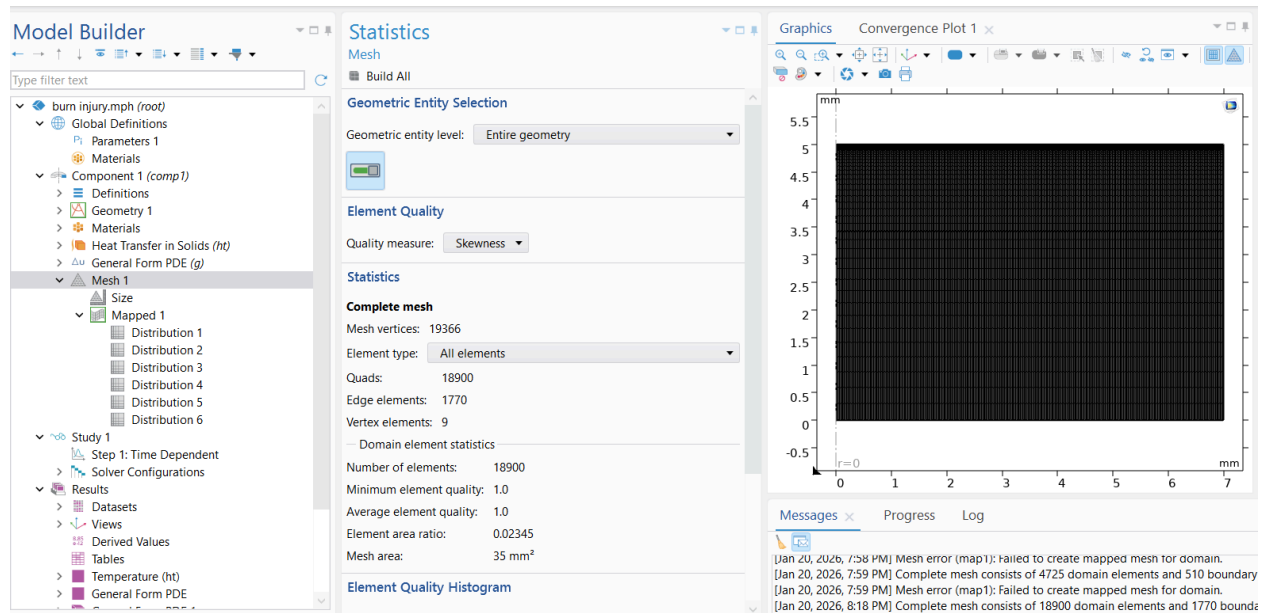
burn injury profile and temperature contour would be:

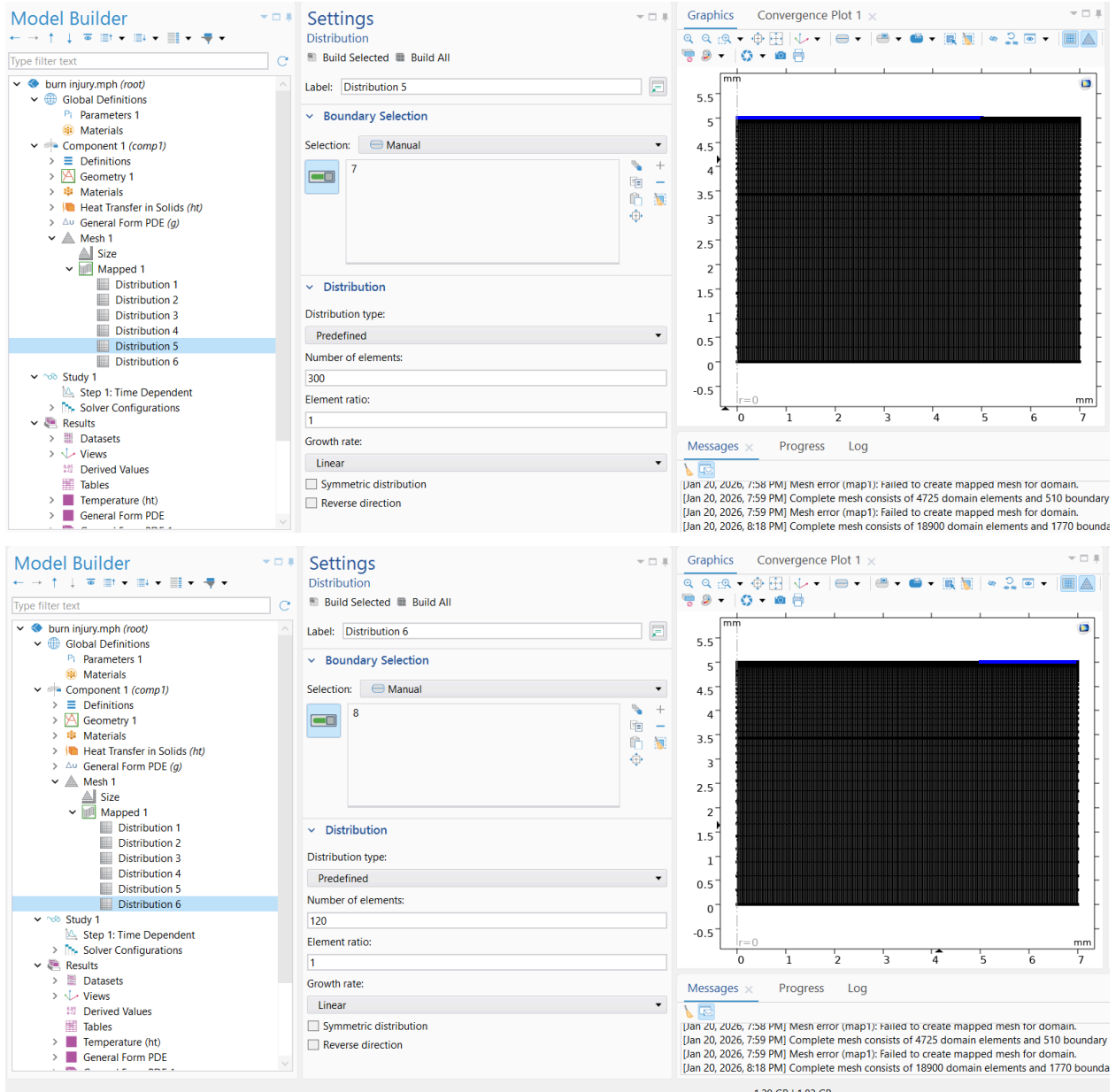




A lower element count results in larger grid cells, which often produces a 'stair-step' or pixelated visualization. While a coarse mesh significantly reduces computation time, it lacks the resolution necessary to capture localized temperature spikes or the precise depth of tissue damage. This loss of spatial fidelity ultimately compromises the accuracy of the boundary interfaces between skin layers

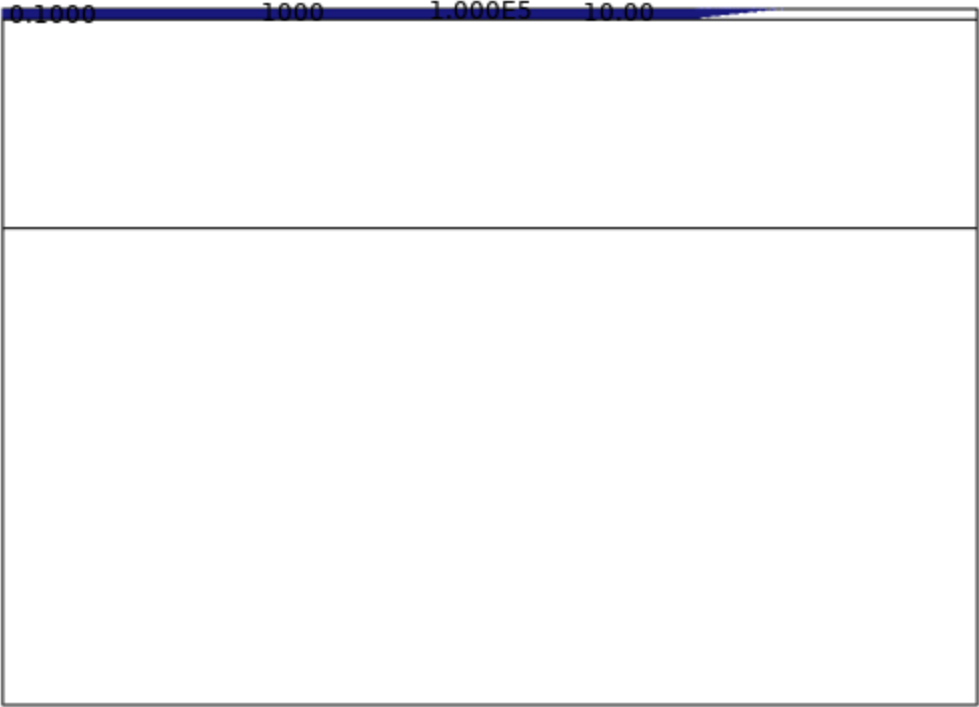
For double base: we take distribution 2, distribution 5, distribution 6 respectively 420, 300 and 120. Thus the total element will be 18900

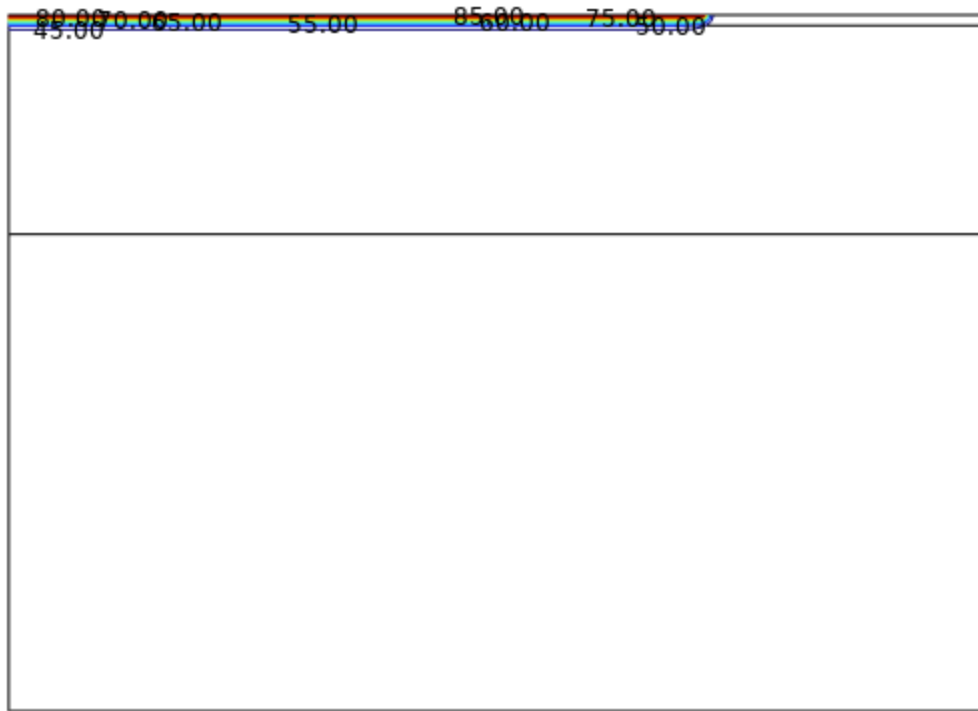




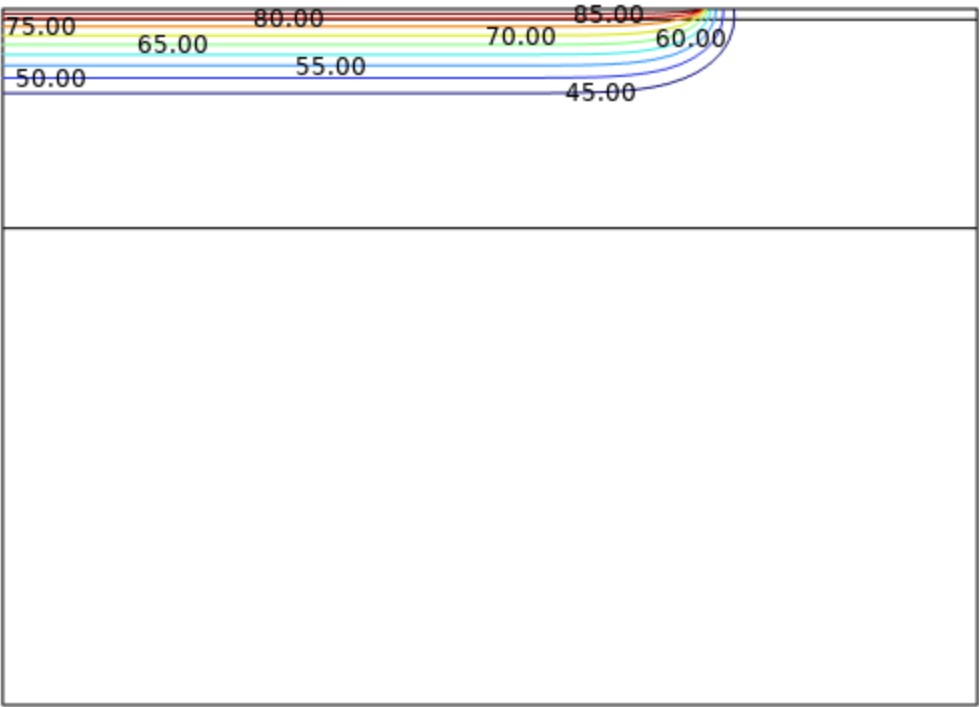
3. Perform the whole simulation for step size, $\Delta t = 1s$, $\Delta t = 0.05s$ and $\Delta t = 0.1s$. For all cases, show burn injury profile and temperature contour. Comment on the changes.

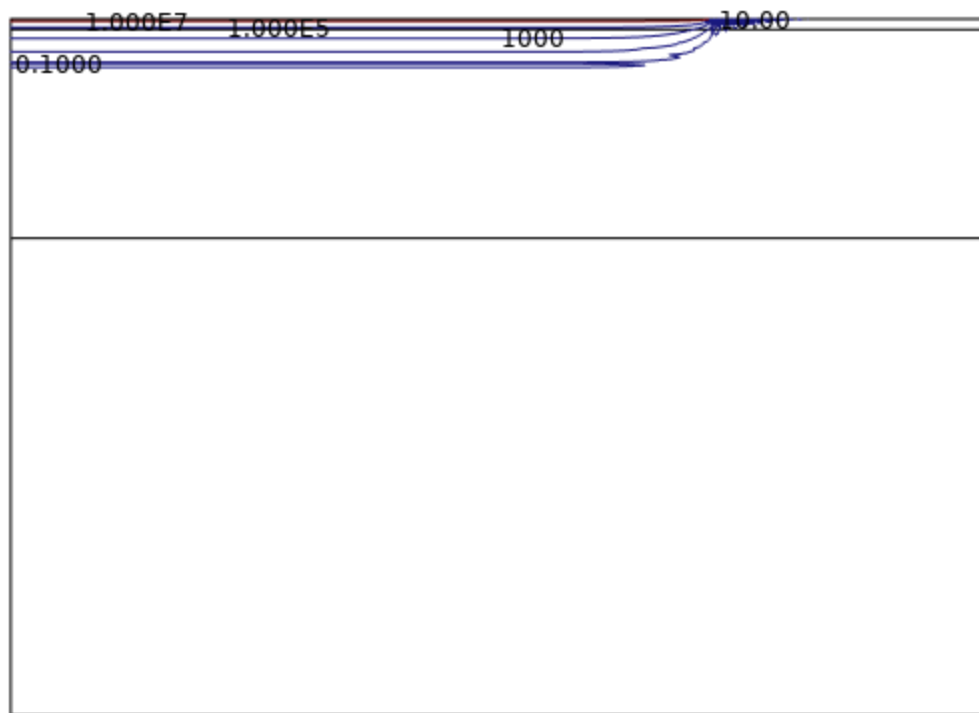
for $\Delta t = 0.05\text{s}$,





for $\Delta t = 1\text{s}$





for $\Delta t = 0.1s$



